Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | **Paper -2(Adv-2024-P2-Model)** | **Date: 24-11-2025**

Time: 02.00Pm to 05.00Pm | **GTA-06** | **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**A**	3	**B**	4	**C**	5	**ABCD**	6	**AB**
7	**ACD**	8	**13**	9	**6**	10	**25**	11	**3**	12	**9**
13	**9**	14	**415**	15	**1520**	16	**26**	17	**509**		

PHYSICS

18	**C**	19	**D**	20	**A**	21	**B**	22	**ABCD**	23	**CD**
24	**ACD**	25	**4**	26	**3**	27	**4**	28	**4**	29	**8**
30	**4**	31	**4**	32	**200**	33	**40**	34	**204**		

CHEMISTRY

35	**A**	36	**B**	37	**C**	38	**A**	39	**ABC**	40	**AD**
41	**AB**	42	**5**	43	**2**	44	**3**	45	**3**	46	**5**
47	**6**	48	**5**	49	**2**	50	**0.1 to 0.2**	51	**0.2 to 0.3**		

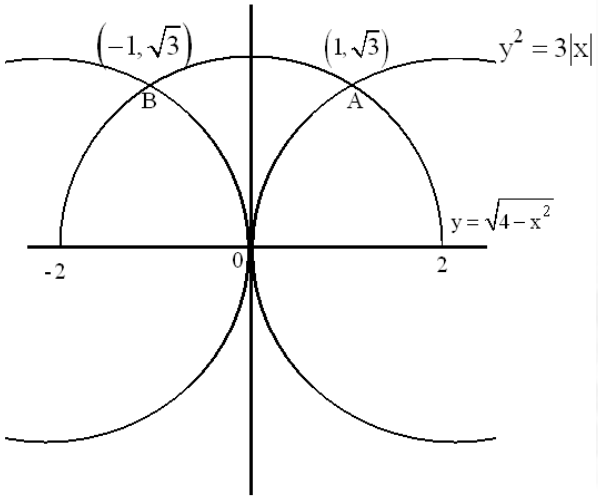
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SOLUTIONS
MATHEMATICS

1. Required area $= 2\int_0^1 \left(\sqrt{4-x^2} - \sqrt{3x}\right)dx$

$$= 2\left(\frac{x}{2}\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right) - \frac{\sqrt{3}.2x^{3/2}}{3}\right)_0^1$$

$$= \frac{2\pi - \sqrt{3}}{3}$$



2. We have $P(A) = P(7) = \frac{6}{36}m$, $P(B) = P(4) = \frac{3}{36}$

Since equal throws are disregarded,
Hence in each throw A is twice as likely to win as B.
Let $P(B) = p$, $P(A) = 2p$

$$3p = 1 \Rightarrow P = \frac{1}{3}$$

3. On solving, we get $\frac{a\pi}{b} = \frac{13\pi}{7} \Rightarrow 13 + 7 = 20$.

4. We have $\left|\frac{z_1}{2} + \frac{z_2}{3} + \frac{z_1}{4} + \frac{z_4}{5}\right| = \frac{k}{60}\left|z_1 z_2 z_3 z_4\right|\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \frac{1}{z_4}\right|$

Now, $z_1\bar{z}_1 = 2$, $z_2\bar{z}_2 = 3$, $z_3\bar{z}_3 = 4$ and $z_4\bar{z}_4 = 5$

So, $k = \frac{60}{\left|z_1 z_2 z_3 z_4\right|} = \frac{60}{\sqrt{2}\sqrt{3}\sqrt{4}\sqrt{5}} = \sqrt{30} = \left|z_4 z_1 z_2\right|$

Note for objective take $z_1 = \sqrt{2}; z_2 = \sqrt{3}; z_3 = 2; z_4 = \sqrt{5}$ |

5. If T comes in first toss then Mr. B can win in only one case that is TT.

$\Rightarrow$ probability of Mr. B winning $= \frac{1}{4}$

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$\Rightarrow$ Probability of Mr.A winning = $\frac{3}{4}$

Given first toss is head, Mr. A can win is successive tosses are T,HT, HHT, …..

Probability $= \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$

Given first toss is head, Mr.A can win is successive tosses are HT, HHT, HHHT,…..

Probability = $= \frac{\frac{1}{4}}{1-\frac{1}{2}} = \frac{1}{2}$

6. Equation of a plane passing through the line $3x-y+2z-1=0=x+2y-z-2$ is $3x-y+2z-1+\lambda(x+2y-z-2)=0$

Since it is perpendicualr to the given plane

$\therefore \lambda = -\frac{3}{2}$

$\therefore$ Equation of the line of projection is

$3x-8y+7z+4=0=3x+2y+z$

Its direction ratios are < 11, – 9 , –15 > and the point $(-1,1,1)$ lies on the line

$\therefore \frac{x+1}{11} = \frac{y-1}{-9} = \frac{z-1}{-15}$ is also the equation of the line of projection.

7. When $f(x)$ is continuous at x = 2 $f'(x)$ *DNE at* $x=2$

and $f'(x)$ changes sign from + to –

$\Rightarrow f(x)$ attains max. At x = 2 if $\frac{k^3(k-1)^2}{k^2-k-2}=0 \Rightarrow k=0,1$

When $f(x)$ is discontinuous at x = 2, $f'(x)$ changes its sign from + to –.

$f(x)$ will attain maximum if $\lim_{x\to 2^+} f(x) < f(2)$ as

$\lim_{x\to 2^-} f(x) = f(2)$ *i.e if* $k \in (-\infty,-1)\cup(0,1)\cup(1,2)$

$\Rightarrow k \in (-\infty,-1)\cup[0,2)$

8. O $\rightarrow$ outside the triangle ABC

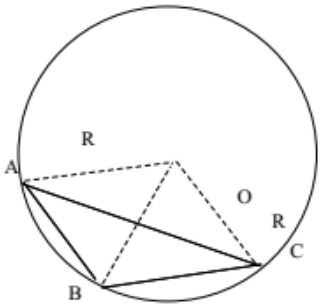
H $\rightarrow$ outside the triangle ABC

$\Rightarrow$ mid point of OH lies on BC

ABC is obtuse angled triangle

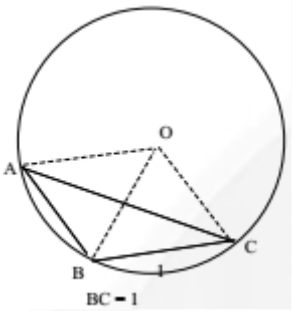
Consider $O(\bar{O})$, $A(\bar{a})$, $B(\bar{b})$, $C(\bar{c})$

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mid point of OH $\frac{\bar{a}+\bar{b}+\bar{c}}{2}$

$\Rightarrow \left|\bar{a}\times\bar{b}-\bar{a}\times\bar{c}\right|=0$



sine rule in Δle BOC : $\frac{R}{\sin\theta}=\frac{BC}{\sin(\pi-2\theta)}$

$1+\sin\theta=\frac{25}{4R^2}\Rightarrow\sin\theta=\frac{25-4R^2}{4R^2}$

but $\cos\theta=\frac{1}{2R}$ and $\sin^2\theta+\cos^2\theta=1$

$\Rightarrow R=\frac{25}{14},\ \cos\theta=\frac{7}{25},\ \sin\theta=\frac{24}{25}$

area of Δle $ABC=\frac{1}{2}R^2\sin\theta+\frac{1}{2}R^2\sin2\theta-\frac{1}{2}R^2\sin(\pi-\theta)$

$=\frac{1}{2}R^2\sin2\theta=\frac{1}{2}\left(\frac{25}{14}\right)^2\left(2\left(\frac{24}{25}\right)\left(\frac{7}{25}\right)\right)=\frac{6}{7}$

9. $f(x)=\operatorname{sgn}\left(\left(x^2-ax+1\right)\left(bx^2-2bx+1\right)\right)$

$P(x)Q(x)=0$ at exactly one value of 'x'

case (i) $\Delta_1=0$ and $\Delta_2=0$

$a^2=4$ and $4b^2-4b<0\Rightarrow b\in(0,1)$

case (ii) $\Delta_1=0$ and $\Delta_2=0$

$a^2-4=0$ $4b^2-4b=0$

a = 2, -2 b = 0, 1

b = 0 (2, 0) (-2, 0)

b = 1 (2, 1) (-2, 1) (not possible)

Total number of ordered pairs (a, b) = 3 + 3 = 6

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10. $\sum_{K=0}^{\infty}\frac{(-1)^K}{(K+1)(2K+1)3^K}=\sum_{K=0}^{\infty}\frac{(-1)^K}{3^K}\cdot\left(\frac{2}{2K+1}-\frac{1}{K+1}\right)$

$=2\sum_{K=0}^{\infty}\left(\frac{-1}{3}\right)^K\int_0^1 x^{2K}dx-\left(\frac{-1}{3}\right)^K\cdot\int_0^1 x^K dx=2\int_0^1\sum_{K=0}^{\infty}\left(\frac{-x}{3}\right)^{2K}dx-\int_0^1\sum_{K=0}^{\infty}\left(\frac{-x}{3}\right)^K dx$

$=\frac{\pi}{\sqrt{3}}-3\log_e\left(\frac{4}{3}\right)=\frac{\pi}{\sqrt{3}}+3\log_e\left(\frac{3}{4}\right)a=3,\ b=4$

$a^2+b^2=9+16=25$

11. $f(x)=\lim_{n\to\infty}\frac{\log_e(e^n+x^n)}{n}=\begin{cases}1, & x\in(0,e)\\ 1, & x=e\\ \log_e^x, & x>0\end{cases}$

$\text{area}=\int_0^{e^2}f(x)\,dx=\int_0^{e}(1)\,dx+\int_e^{e^2}\log_e^x\,dx\ =(e)+x\ln x-x\Big|_e^{e^2}$

$=(e)+\left(e^2(2)-e^2\right)-(e-e)=e^2+e$ (consider e = 2.718 approximately) $\cong 10.1055$

12. Circles is $(x+2)^2+(y-3)^2=1$

$L_1: x+y-1=0$

$L_2: x-y+4=0$

On solving circles and $L_1=\left(-2-\frac{1}{\sqrt{2}},3+\frac{1}{\sqrt{2}}\right)$

Solving circles and $P(L_2)=\left(-\frac{3}{2},\frac{5}{2}\right)$

$|Z_1|^2+2|Z_2|^2=14+5\sqrt{2}+17=31+5\sqrt{2}$

$\alpha+\beta=36$

13. $R=\lim_{x\to0^+}f(x)=\frac{\pi}{2}$

$L=\lim_{x\to0^-}f(x)=\frac{\pi}{4}$

$\frac{16}{\pi^2}(L^2+R^2)=\frac{16}{\pi^2}\left(\frac{\pi^2}{2}+\frac{\pi^2}{11}\right)=9$

14. $\left(\sqrt{3}+1\right)^6=I+F$ where $0\le F<1$

Let $\left(\sqrt{3}-1\right)^6=G$ where $0\le G<1$

$\therefore I+F+G=\left(\sqrt{3}+1\right)^6+\left(\sqrt{3}-1\right)^6$

$=2\left\{{}^6C_0\left(\sqrt{3}\right)^6+{}^6C_2\left(\sqrt{3}\right)^4+{}^6C_4\left(\sqrt{3}\right)^2+{}^6C_6\right\}$

I = 416 – (F + G)

but $0\le F+G<2$ and $F+G$ has to be an integer $\therefore I=416-1=415$

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15. The general term in the expansion of $\left(1+x+x^2\right)$ is $\frac{20!}{r!s!t!}1^r x^s x^{2t}$

Where $r + s + t = 20$

For coefficient of x^3, $s+2t=3 \Rightarrow t=0, s=3, r=17$

Or $t = 1, s = 1, r = 18$

$\therefore$ coefficient of $x^3 = \frac{20!}{17!3!}+\frac{20!}{18!1!}=\frac{4.20!}{18!}$

16 & 17.

$P^TP=I$

$P_K=P^T\left(PAP^T\right)\left(PAP^T\right).....\left(PAP^T\right)P$

$=\left(P^TP\right)A\left(P^TP\right).....A\left(P^TP\right)$

$(I)A(I)A.....A(I)\underbrace{AA.....A}_{\text{k times}}=A^K$

Similarly $T_K=B^K$

$T_K=B^K$

$A^2=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}=\begin{bmatrix}5 & -8\\ 2 & -3\end{bmatrix}$

$B^2=\begin{bmatrix}\frac{1}{2} & 3\\ 0 & 0\end{bmatrix}=\begin{bmatrix}\frac{1}{2^2} & \frac{9}{2}\\ 0 & 1\end{bmatrix}$

A) $\sum_{K=1}^{5}a_K=a_1+a_2+.....+a_5=3+5+7+9+11=35$

B) $\sum_{K=1}^{3}b_K=b_1+b_2+b_3=-1-3-5=-9$

C) $\sum_{K=1}^{\infty}X_K=X_1+X_2+.....\infty=\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....+\infty=\frac{\frac{1}{2}}{1-\frac{1}{2}}=1$

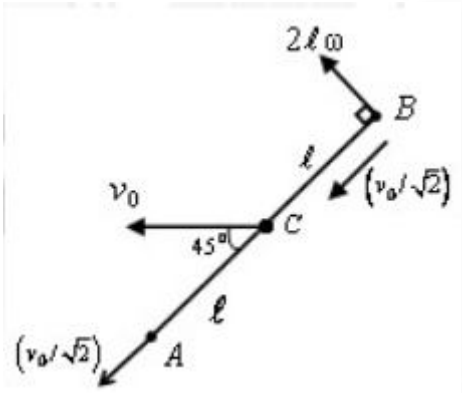
D) $\sum_{K=1}^{10}Y_K=Y_1+Y_2+.....+Y_1=\underbrace{1+1+.....+1}_{\text{10 times}}=10$

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PHYSICS

18. Angular velocity of rod is $\omega = \dfrac{v_0 \sin 45}{l}$

From figure $v_B = \sqrt{\left(\dfrac{v_0}{\sqrt{2}}\right)^2 + (2l\omega)^2} = \left(\sqrt{\dfrac{5}{2}}\right)v_0$



19. $2\mu d = (2n+1)\lambda/2$ for constructive interference $\lambda = \dfrac{4\mu d}{2n+1}$

$\lambda = 4\mu d, \dfrac{4\mu d}{3}, \dfrac{4\mu d}{5}, \dfrac{4\mu d}{7}$.....As $n \uparrow \lambda \downarrow$

$400 = \dfrac{4\mu a}{2n+1} \Rightarrow 4\mu d = 400(2n+1).....(1)$

again $\lambda = \dfrac{4\mu d}{2(n-1)+1}$ for next one $\lambda < 1150 \Rightarrow \dfrac{4\mu d}{(2n+1)-2} < 1150$

$2n+1 > \dfrac{230}{75} n > \dfrac{31}{30} \therefore n = 2, 3, 4.....$

$\Delta L = 2lat \Rightarrow T = \dfrac{d}{hd} = 3.125^0\,C$ and $\lambda = \dfrac{4\mu d}{2(2-1)+1} = \dfrac{4\times 1\times 500}{3} = 666.67$ nm

20. At the time of collision, the line joining the centers is along the line QD at an angle 45^0. The Distance between the centers is3r(r + 2r). Required distance

$= \dfrac{L}{2} + 3r\sin 45^0 = \dfrac{L}{2} + \dfrac{3r}{\sqrt{2}}$

21. It behaves as two current carrying sheets carrying currents in the opposite directions. At P the fields are in the same direction and at Q they are in the opposite directions.

Ampere's law gives $B = \dfrac{1}{2}\mu_0 k$; k = current per unit width

$k = 1\times v\pi = \sigma[r\omega] \int_0^{\omega} d\omega = \int_0^{t} (at^2 + bt + c)dt \;\; v = r\omega$

22. When only S_2 is open, $\dfrac{2}{3} = \left(\dfrac{4}{120}\right)l \Rightarrow l = \dfrac{2}{3}\times\dfrac{120}{4} = 20$ cm

When only S_1 is open, $2 = \dfrac{4}{120}\times l \Rightarrow l = 60$ cm

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When all three are closed, $2=\frac{4}{120}\times l\Rightarrow l=60\text{ cm}$

When only S_2 is open $2=\frac{4}{120}\times l\Rightarrow l=60\text{ cm}$

23. $q_{net}f=\sigma\left(T_A^4-T_1^4\right)=\sigma\left(T_{n-1}^4-T_n^4-T_B^4\right)$

$\Rightarrow\sigma\left(T_A^4-T_1^4\right)=(n+1)q_{net}f\Rightarrow q_{net}f=\frac{\sigma}{(nH)}\left(T_A^4-T_B^4\right)\Rightarrow\frac{q_{net.f}}{q_{net.f}}=\frac{1}{n+1}$

24. $\tan i=\frac{x}{y}$, $x=y\tan i$, $dx=y\sec^2 i\,di$. $\tan r=\frac{x'}{y'}$

$x'=y'\tan r$. $dx=y'\sec^2 r\,dr$. $dx=dx'\Rightarrow y\sec^2 i\,di=y'\sec^2 r\,dr$

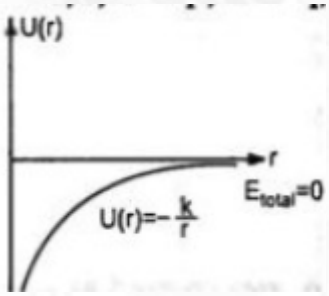
$y'=y\frac{\sec^2 i}{\sec^2 r}\frac{di}{dr}=y\frac{\cos^2 r}{\cos^2 i}\frac{di}{dr}.....(1)$

From snell's law :

so from eq.(i) $y'=\frac{y}{\mu}\left(\frac{\cos^3 r}{\cos^3 i}\right)$ for $(P)\,y'=\frac{8}{4}\left(\frac{\cos^3 53^0}{\cos^3 37^0}\right)=2.53$

for $(Q)\,x'-x=y'\tan r-y\tan i$

for $(S)\,P_{out}=\frac{P}{4\pi}2\pi(1-\cos C)=\frac{P}{2}\left(1-\frac{\sqrt{7}}{4}\right)\frac{P_{out}}{P}\times100=50\left(1-\frac{\sqrt{7}}{4}\right)$



25. $P_1=\frac{4T}{a}$, $V_1=\frac{4}{3}\pi a^3$

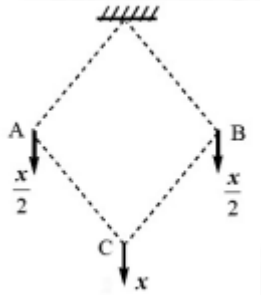
$P_2=\frac{\sigma^2}{2\varepsilon_0}=\frac{4T}{2a}$, $V_2=\frac{4}{3}\pi(2a)^3$ $\because$ process is adiabatic,

$P_1V_1^{5/3}=P_2V_2^{5/3}$ solving we get $\sigma=\sqrt{\frac{15\varepsilon_0 T}{4a}}$

26. $v_{rms}^2=\frac{\int v^2dn}{\int dN}=\frac{\int v^2\frac{dN}{dv}dv}{\int\frac{dN}{dv}dv}=\frac{\int_0^3 v^2\left(-\frac{5}{3}v+5\right)dv}{\int_0^3\left(-\frac{5}{3}v+5\right)dy}=1.50$

27. In point 'C' is displaced by 'x' then point A and B both moves by x/2 additional deformation in each spring is $\frac{x}{2\sqrt{2}}$ restoring force on block $\frac{kx}{2\sqrt{2}}\frac{1}{\sqrt{2}}2=mx$

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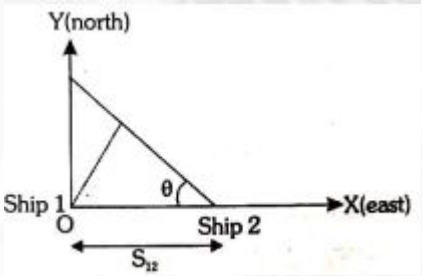
$\frac{k}{2}(x)=mx \;\; T=2\pi\sqrt{\frac{2m}{k}} \;\; x=4$

28. $I=I_0\sin^2\theta$

$dI=I_0(2\sin\theta\cos\theta)\,d\theta$

$\frac{d\theta}{\theta}\times100=\left(\frac{dI}{I_0\sin2\theta}\right)\left(\frac{6}{\pi}\right)\times100=\frac{4\sqrt{3}}{\pi}\times10^{-2} \;\; n=4$

29. $\vec{v}_{21}$ (relative velocity of 2 in relative to1) $\vec{v}_{21}=\vec{v}_2-\vec{v}_1=40\hat{j}-30\hat{i}$

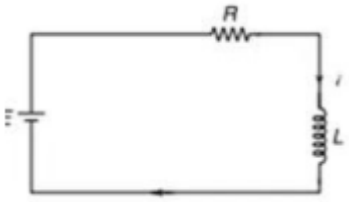


From the concept of relative velocity 1 may be assumed to be fixed and 2 to be moving with the velocity $\left(-30\hat{i}+40\hat{j}\right)$. This is a vector of magnitude 50 and inclined at $\theta=\tan^{-1}\left(\frac{4}{3}\right)$. The shortest distance is obviously the perpendicular distance of the relative velocity vector from the fixed position of ship 1.

$\therefore S_{shortest}=s_{12}\sin\theta$ where s_{12} is the initial distance between 1 and 2.

$s_{12}=10\times(4/5)=8$ km 00

30. Just before closing S_2, the effective circuit is as shown.

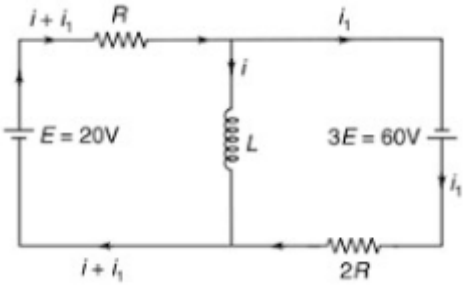


$i=\frac{E}{R}=\frac{20}{10}=2A$

Immediately, after closing S_2, the current through inductor will remain I = 2A. The effective circuit has been shown in figure.

Applying KVL to the bigger loop

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$(i+i_1)R+i_1(2R)=4E \qquad \Rightarrow 3i_1R=4E-iR \Rightarrow i_1=2A$

Negative sign says that current through the inductor is decreasing energy stored in magnetic field. $\dfrac{dU_B}{dt}=Li\dfrac{di}{dt}=0.5\times2\times(-40)=-40\ J/s$

Negative sing says that energy stored in the inductor is decreasing.

31 & 32

$\theta=\dfrac{\omega}{R}=\dfrac{1.6}{100}=\dfrac{y}{2}\ y=\dfrac{1.6}{100}\times1\ mA \quad y=16\ mm \qquad \lambda=\dfrac{h}{mv}=6.625\times10^{11}$

For central point to be minimum

$y=(2n+1)\dfrac{\lambda D}{2d}\ d=\dfrac{(2n+1)\lambda D}{2y}$

$\therefore d_m=\dfrac{6.625\times10^{-11}\times1}{2\times1.6\times10^{-2}} \quad d_m=\dfrac{6.625}{3.2}\times10^{-9}\ mA \ \therefore d_m=2.07\ nm$

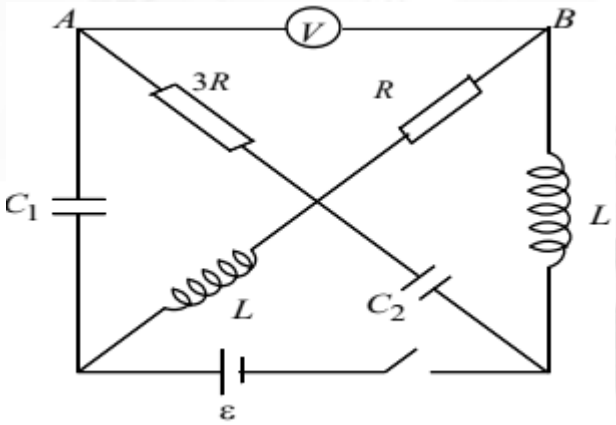
33. Before switch is close

Change on capacitor $C_2=Ce$

Change on capacitor $C_1=0$

When switch is open initially $i=\dfrac{\varepsilon}{R}$

Reading of voltmeter is $\dfrac{\varepsilon\times3R}{R}-\dfrac{\varepsilon\times R}{R}=2\varepsilon$



34. There is no battery in circuit and there is no any L – C oscillation type circuit, so all energy is dispatted in heat loss

$\varepsilon=\dfrac{1}{2}C\varepsilon^2+2\times\dfrac{1}{2}Ci^2=\dfrac{1}{2}C\varepsilon^2+L\left(\dfrac{\varepsilon}{R}\right)^2=\dfrac{1}{2}C\varepsilon^2+\dfrac{L\varepsilon^2}{R^2}$

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CHEMISTRY

35. $z = 1 + \frac{Pb}{RT}(\text{high pressure})\ y = c + mx$

36. It cannot give Cl_2

37. T_c^{3+} ion contains 4 unpaired electrons

38. Solvay's process

39. 1 lit of acidified $KMnO_4$ contain = 15.8 g

$$FeS_2 + MnO_4^- \xrightarrow{H^+} Fe^{3+} + 2SO_2 + Mn^{2+}$$

Also $KMnO_4 + SO_2 \xrightarrow{H^+} Mn^{2+} + SO_4^{2-}$

Eq. of $KMnO_4$ = eq.of SO_2

Or $0.1 \times 5 = \text{mole of } SO_2 \times 2 \Rightarrow \text{mole of } SO_2 = \frac{0.5}{2} = 0.25$

Volume of SO_2 at STP = $0.25 \times 22.4 = 5.6\text{lit}$.

Mole of FeS_2 used = $\frac{1}{2} \times \text{mole of } SO_2$ produced $= \frac{0.25}{2} = 0.125$

40. Conceptual

41. Option C gives hydrazine and option D gives ammonia

42. $\Delta T_f = 2K_f m$

$0.558 = i \times 1.86 \times 0.1 \Rightarrow i = 3$

$i = 1 + (n-1)\alpha \Rightarrow 3 = 1 + (n-1)1 \Rightarrow n = 3$

Thus the complex is $\left[Co(NH_3)_5 Cl\right]Cl_2$

43. $t_{1/2} = 20\,\text{min},\ t'_{1/2} = 10\,\text{min}\ \&\ [A]'_0 = 2[A]_0$

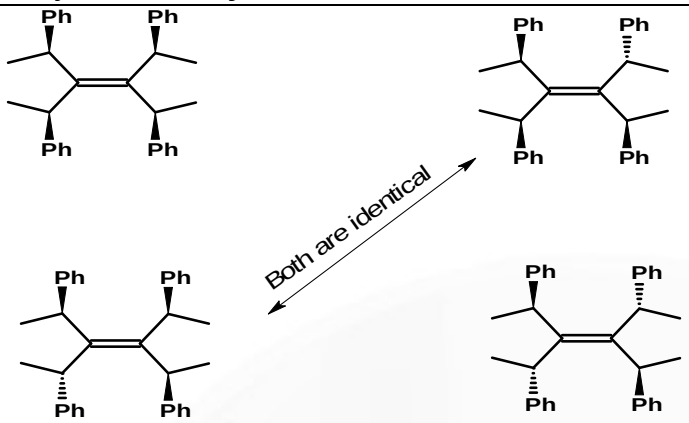
$$\because t_{1/2} \alpha \frac{1}{[A]_0^{n-1}} \Rightarrow \frac{t'_{1/2}}{t_{1/2}} = \left[\frac{[A]'_0}{[A]_0}\right]^{n-1}$$

$$\text{or } \frac{20}{10} = \left[\frac{2 \times [A]_0^{n-1}}{[A_0]}\right]^{n-1} \rightarrow 2 = 2^{n-1} \rightarrow n-1 = 1 \Rightarrow n = 2$$

44. Conceptual

45.

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46. Ir^{2+} in square planner complex contains five non axial and two axial electrons

47. N – Co – O $\left(90^0\right)$ angles are equal to 6

48. Conceptual

49. Conceptual

50. $25 \times N_1 = 40 \times 0.1$

51. % of strength = $1.7 \times$ normality

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